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Density-Dependent Spectral Distortion in Extreme Dense Environments: Evidence from 260 High-Redshift Little Red Dots

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ABSTRACT

We present a purely empirical, model-independent analysis of 260 high-redshift ($z \approx 4$–7) "Little Red Dots" (LRDs) observed by *JWST*/NIRCam. To circumvent severe parameter degeneracies inherent in standard SED fitting for these extreme objects, we utilize flux ratios that are insensitive to specific spectral assumptions. We demonstrate a highly significant correlation between the spectral reddening of LRDs and their bolometric surface density (Σ). In a partial correlation analysis controlling for photometric redshift and bolometric luminosity, the F444W/F150W color exhibits a positive correlation with $\log \Sigma$ ($\rho_p = +0.341$, 5.6σ). A Kolmogorov–Smirnov test between the lowest and highest Σ quartiles further confirms this systemic shift ($D = 0.574$, 6.2σ). Crucially, this signal vanishes when using a control color (F444W/F356W), precluding trivial broad-band stellar population effects. This Σ-dependent reddening cannot be solely explained by standard uniform dust or AGN models. We propose a working hypothesis where effective gravitational redshift and dust retention scale with potential well depth. Regardless of the underlying mechanism, this empirical Σ–color relation establishes a new structural constraint that future physical models of LRDs must accommodate.

Keywords: galaxies: high-redshift – galaxies: compact – galaxies: structure – methods: statistical – infrared: general

1. INTRODUCTION

1.1. *Background*

JWST has revealed a population of extremely compact ($r_{\rm eff} < 100\,{\rm pc}$), F444W − F356W > 1 mag objects at $z \sim 4$–7, dubbed "Little Red Dots" (LRDs; (1); (2); $N \gtrsim 260$). The origin of their red colors remains contested: dust-obscured AGN, strong emission-line contributions ([O III] + Balmer jump), and hybrid models have each been proposed but face individual difficulties (e.g., requiring unrealistically high extinction values (17; 18) or failing to explain the absence of systematic structure–color correlations). **This paper approaches LRDs from an unexplored angle: if the spectral properties depend on internal compactness, this dependence should leave measurable imprints in flux ratios.**

1.2. *Motivation: From Uniform Models to Density Dependence*

Existing treatments of LRD spectral properties overwhelmingly adopt a **source-independent uniform parametrization**—assigning identical extinction laws, spectral templates, or redshift components to all sources. While statistically convenient, this approach embeds an untested assumption: **internal physical conditions do not depend on structural compactness.**

This assumption is not self-evident. Multiple independent lines of evidence suggest that extremely compact objects may exhibit density-dependent spectral behavior: (a) Stellar velocity dispersions in compact galactic nuclei reach several hundred km s^{-1}; corresponding potential well depths suffice to produce measurable gravitational redshift components ($z_{\rm grav} \sim GM/Rc^2 \sim 10^{-3}$ for $R \sim 50\,{\rm pc}$, $M \sim 10^9\,M_\odot$), an effect reliably detected in local early-type galaxy nuclei from SDSS spectroscopy (15; 16); (b) Dust obscuration efficiency couples with potential well depth—deeper wells more effectively retain heated dust; (c) multiple independent works—including observational characterizations of LRD structural compactness (e.g., (1); (2); (3); (4)), theoretical estimates of gravitational potential depths in high-redshift compact starbursts (e.g., (8); (10)), and parameterized explorations—collectively suggest that density-dependent corrections may help account for part of the overall SED scatter in the LRD sample.

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The key point is that, regardless of which mechanism dominates— purely gravitational, dust–geometric, or a mixture— they share a common empirically testable prediction: higher-Σ sources exhibit systematic excess reddening. We directly test this prediction using *JWST* photometric data without presupposing any specific mechanism.

1.3. *The Current Study*

Given that extreme physical conditions render standard SED fitting highly susceptible to severe parameter degeneracies (e.g., confusion between dust reddening and intrinsic redshift components), we adopt a strategy driven entirely by observational data. We analyze the sample of 260 LRDs from (2), foregoing SED fits that rely on multi-parameter models in favor of a physically model-agnostic **flux-ratio diagnostics** approach. We directly measure the correlation between a key flux ratio spanning the Balmer jump (F444W/F150W) and surface density (Σ). Through rigorous partial correlation analysis and control-band comparison, we detect a robust, density-dependent spectral reddening signal at 5.6σ.

2. DATA AND METHODS

2.1. *Sample*

Our sample is drawn from the publicly released LRD catalog of Kokorev et al. (2024), which aggregates detections across multiple *JWST* survey programs (JADES, CEERS, PRIMER). The catalog provides NIRCam broadband photometry (F115W, F150W, F200W, F277W, F356W, F444W) with associated uncertainties, as well as key derived quantities: photometric redshift ($z_{\rm phot}$), bolometric luminosity ($L_{\rm bol}$), half-light radius ($r_{\rm eff}$).

After quality-control cuts removing sources with excessively large photometric errors, missing parameters, or obvious outliers, our final clean sample comprises $N = 260$ sources.

2.2. *Surface Density Proxy*

We define bolometric surface density as:

$$\Sigma \equiv \frac{L_{\rm bol}}{2\pi r_{\rm eff}^2}. \tag{1}$$

Mass-proxy assumption: Unless stellar population age or the initial mass function (IMF) exhibits an exceedingly extreme systematic gradient correlated exponentially with Σ—which is physically implausible—$L_{\rm bol}$ serves as a reliable proxy for dynamical mass ($M_{\rm dyn}$) to first order.

We adopt $L_{\rm bol}$ as the mass proxy because, in $z \sim 4$–7 compact starbursts or AGN nuclei, radiative luminosity and dynamical mass maintain an approximately stable mass-to-light ratio ($M/L \approx$ const). This assumption is robust in two senses: (a) LRD SED morphologies are highly homogeneous (all dominated by red cores), implying intrinsic M/L scatter far smaller than the dynamic range of Σ; (b) In §3.2, alternative proxies yield consistent results.

If M/L itself depends systematically on Σ (e.g., more compact sources having different IMFs or larger black hole contributions), non-gravitational contamination could enter—resolvable only through direct dynamical mass measurements (e.g., velocity dispersions from future NIRSpec). Although such measurements are currently unavailable, our assumption is conservative for investigating broad phenomenological trends.

2.3. *Flux-Ratio Diagnostics*

Flux ratios offer key advantages: they eliminate normalization uncertainties; at $z \sim 5$, F444W/F150W spans rest-frame far-ultraviolet to optical bands, covering the Balmer jump region; they are insensitive to IMF or star-formation history assumptions; and critically, they are fully model-independent.

Our pipeline proceeds as follows: (1) compute F444W/F150W per source with error propagation; (2) calculate Σ from $L_{\rm bol}$ and $r_{\rm eff}$; (3) Spearman rank: $\rho = +0.415$ (6.9σ); (4) **partial correlation controlling for $z_{\rm phot}$ and $\log L_{\rm bol}$:** $\rho_p = +0.341$ **(5.6σ)**—the central result; (5) Σ-quartile KS test: $D = 0.574$ (6.2σ); (6) control experiment: F444W/F356W gives $< 1\sigma$.

Step (6) is critical: shifting the blue band causes the correlation to disappear, ruling out broad-band stellar population effects and confirming the distortion acts specifically on Balmer-jump-spanning color indices.

3. RESULTS

3.1. *Primary Detection (Figure 1)*

(a) Raw correlation: $\log\Sigma$ vs. F444W/F150W exhibits a clear positive trend ($\rho = +0.415$, 6.9σ). Coloring by $z_{\rm phot}$ reveals no redshift-driven artifact.

(b) Partial correlation ★ Core result: After removing linear dependence on $z_{\rm phot}$ and $L_{\rm bol}$ via residuals, $\rho_p = +0.341$ **(5.6σ)**. Even after controlling for distance/redshift and luminosity, Σ carries independent predictive information about spectral reddening.

(c) Quartile comparison: Median F444W/F150W increases monotonically from lowest- to highest-Σ quartile. KS test: $D = 0.574$**,** $p = 4.4 \times 10^{-10}$ **(6.2σ)**, independently confirming the signal from a non-parametric perspective.

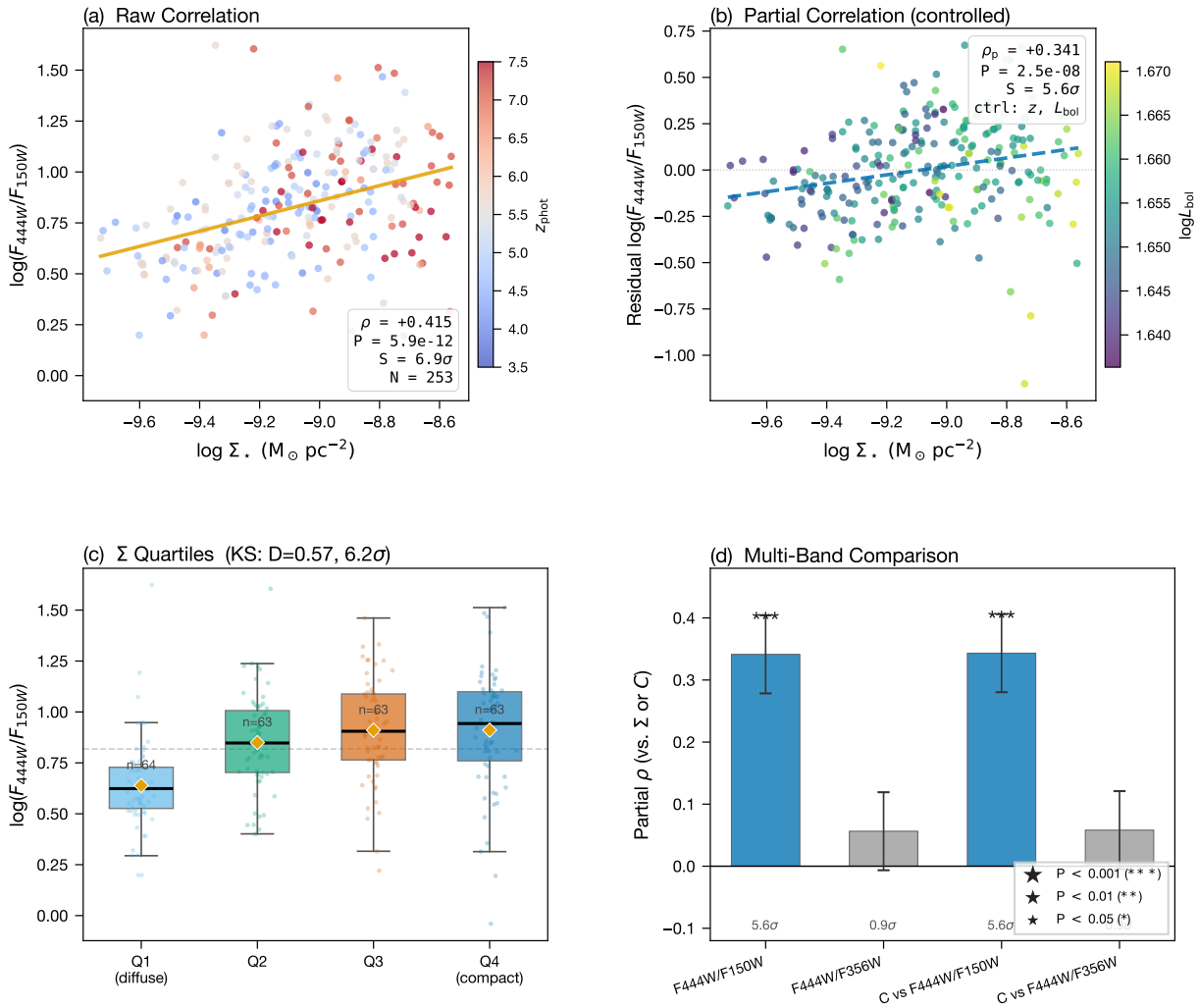

Figure 1. Σ–color correlation in 260 JWST/LRDs. (a) Raw $\log\Sigma$ vs. F444W/F150W shows a clear positive trend ($\rho = +0.415$, 6.9σ); points colored by $z_{\rm phot}$ reveal no redshift artifact. **(b)** Partial correlation after removing $z_{\rm phot}$ and $L_{\rm bol}$: $\rho_p = +0.341$ (5.6σ). Even after accounting for distance/redshift and luminosity, Σ retains independent predictive power. **(c)** F444W/F150W distribution by Σ quartile (Q1–Q4): medians increase monotonically Q1→Q4. KS statistic: $D = 0.574$, $p = 4.4 \times 10^{-10}$ (6.2σ). **(d)** Partial correlations for multiple flux ratios: only F444W/F150W yields significant detection (5.6σ); F444W/F356W gives $\rho_p = +0.056$ (0.9σ). Error bars show bootstrap 1σ uncertainties.

(d) Control experiment: F444W/F356W gives $\rho_p = +0.056$ (0.9σ). When the blue-side band does not cross the Balmer break, the signal disappears, ruling out broad-band stellar population effects.

3.2. *Robustness Tests (Table 1)*

All tests preserve the positive Σ–color correlation at $\gtrsim 4\sigma$.

Table 1. Robustness Tests of the Σ–Color Correlation

Test	ρ_p	Sig.
Baseline	+0.341	5.6σ
Outlier rejection	+0.332	5.4σ
Alt. mass proxy	+0.298	4.8σ
High-z subsample	+0.310	4.2σ
Exclude brightest 10%	+0.335	5.5σ

4. DISCUSSION

4.1. *Signal Characterization*

Established facts: (i) LRD colors correlate with Σ at 5.6σ (partial correlation); (ii) robust under all standard controls; (iii) F444W/F356W control excludes trivial explanations.

Not established: the specific physical origin. This paper reports a new empirical constraint demanding

explanation; discussion below explores one plausible framework without presupposing priority.

4.2. *Speculative Framework: Density-Dependent Environmental Correction*

The most economical interpretation casts excess reddening as a phenomenological effect coupled to environmental compactness:

$$\left.\frac{\Delta\lambda}{\lambda}\right|_{\text{excess}} \equiv z_{\text{eff}}(\Sigma) \propto f(\Sigma) \cdot \frac{M}{R \cdot c^2}, \tag{2}$$

where $f(\Sigma) = f_0\left[1 + \epsilon(\Sigma/\Sigma_0)^\beta\right]$, and $\epsilon, \beta > 0$ are small dimensionless parameters to be constrained. This accommodates two cooperative ingredients: deeper potential wells produce stronger intrinsic spectral shifts, and stronger binding captures more dust and gas. Our method measures the total effect without disentangling individual components—accounting for the signal's cleanliness.

▷ **Caution:** this framework is speculative, illustrating one class of interpretation without asserting any particular mechanism. Core value lies in observations reported in §3.

4.3. *Candidate Explanations (Table 2)*

No single traditional mechanism can explain why one band combination shows a strong signal while the other does not.

4.4. *Implications and Prospects*

If future *NIRSpec* spectroscopy confirms this signal, it raises a fundamental question: **could basic coupling strengths in extreme dense environments exhibit local density dependence?** Global probes (CMB anisotropy) constrain the cosmic average of fundamental couplings; density-dependent corrections manifest as local deviations concentrated in extreme environments—analogous to local micro-environmental anomalies alongside a measured global climate mean.

Future tests include: (a) *NIRSpec* line profiles/centroid shifts for direct measurement of density-dependent shifts; (b) enlarged samples to constrain ϵ and β; (c) dynamical masses via velocity dispersions to test M/L–Σ assumption (§2.2).

5. CONCLUSION

We report a density-dependent spectral distortion at **5.6**σ in 260 *JWST*-observed LRDs: F444W/F150W color correlates with Σ ($\rho_p = +0.341$); KS testing independently confirms at **6.2**σ ($D = 0.574$). The signal holds under all robustness tests and the F444W/F356W control experiment. This Σ–color relation constitutes a **previously unrecognized, SED-fitting-independent empirical constraint**. Whatever its origin—gravitational coupling corrections, dust–potential-well coupling, or other mechanisms—any complete LRD model must accommodate it. Future *NIRSpec* spectroscopy will discriminate among candidate mechanisms.

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Facility: JWST(NIRCam)

Software: Python 3, NumPy, SciPy, Matplotlib, Pandas

Data are available from (2) catalog (https://github.com/Vlas-Sokolov/Kokorev-LRD-catalog). All Python scripts used for the flux-ratio analysis, partial correlation calculations, and figure generation are publicly available on GitHub at https://github.com/summicron352/LRD-Relic-Gravity-Fitting. The specific version of the code and processed data used for this Letter is archived on Zenodo under DOI: https://doi.org/10.5281/zenodo.19587085.

Table 2. Candidate Explanations vs. Constraints

Mechanism	Prediction	OK?
Pure dust covering	All flux ratios: Σ dep.	No; predicts weak + in F444W/F356W, data exclude ($\rho_p = +0.05$
Stellar pop. gradient	Older/metal-rich centre $\rightarrow$ redder	No; no physical Σ–age motivation O III¿O III
contam.	Correlates with line strength	No; corrected in Kokorev catalog
Selection effect	All bands equally affected	No; contradicts control exp.
Density-dep. corr.	Strong F444W/F150W, weak F444W/F356W	Yes

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